

**What are the barriers to accessing veterinary healthcare among
marginalized livestock farmers, and how do these impact herd
health and household resilience?**

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Abstract

Livestock agriculture serves as a major source of survival for many marginalized rural households in India because it provides a dependable source of income while simultaneously working as a cushion against unexpected hardships or calamities that may occur. Despite its vibrant significance, access to quality and affordable livestock healthcare to such communities is still a question India is struggling with. This paper seeks to analyze multiple hurdles in the way of marginalized livestock farmers to access veterinarian healthcare and the complexity of the challenges faced along with the dire consequences of these barriers, the role of government in the equation and ultimately, would offer viable suggestions and recommendations to improve the situation. The study would also critically analyze the situation through the example of two preexisting case studies which would reveal how the constraints to accessing veterinarian healthcare affect the entire scenario of livestock agriculture negatively. Eventually, these issues also attack the broader community resilience. Ultimately, improvement in access to veterinarian healthcare for marginalized farmers is not just about safeguarding animal health; it is an extremely important task to strengthen the country's health, wealth and education through securing rural livelihood.

Keywords: Livestock, Farmers, Vaccination, Resilience, Barriers, Consequences, Veterinarian Professionals, Governments, Suggestions.

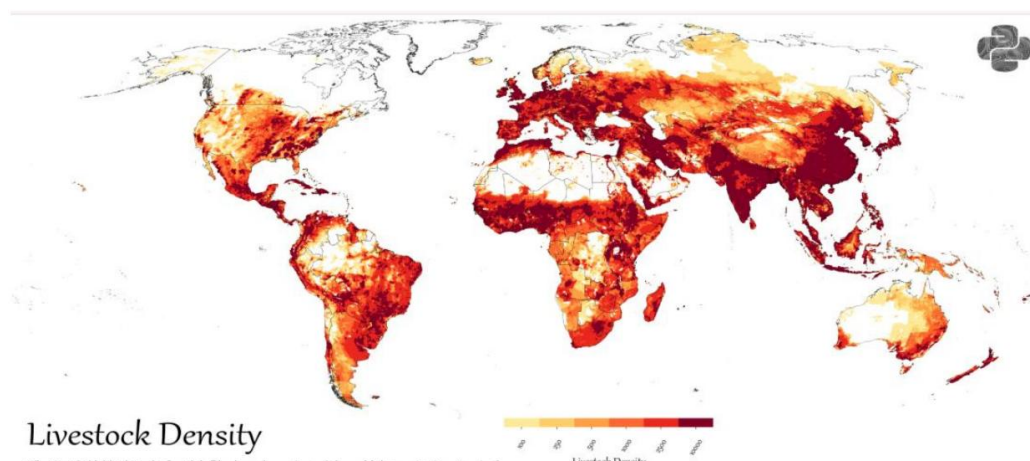
Introduction and Context

Livestock farming refers to breeding and rearing of animals at a designated place (usually called a farm) to gain financial benefits. This is done through either selling their produce like milk, eggs, wool etc or their meat which includes fish, poultry, beef etc. A

healthy livestock industry is often indicative of a country's economic and food stability because it means that the country does not rely on exports and can produce its own food in emergency situations like war and natural calamities (Holden and Garrigus, 2025). Likewise, barriers to livestock farming can be seen as the factors that lead to unavailability or delay in the procurement of medicines or professional help for the animals. This research paper aims to elucidate how these barriers are situated in the practical scenario of livestock industry and how all of this affects the farmers involved, especially those that come from the marginalized sections of the society. Before diving in the subject matter, it is important to understand the basics of This industry. Although the paper will occasionally touch upon the global matrix of the subject, it will primarily focus on the Indian context. Modern Indian history has seen growth and development in nearly all sectors of the economy but owing to its suitable climatic conditions, fertile soil and availability of resources, it has conventionally been regarded as an agrarian-major economy and even today, there is some truth to it. According to a report published by the government of India, agriculture as a sector registered a growth rate of 4.18% between 2019 and 2014, making agriculture contribute nearly 19% in the country's GDP at current prices and giving livelihood to 42.3% population in the country (PIB, 2024). Livestock farming is an ever evolving and extremely substantial part of agriculture in India. Between 2014–15 to 2020–21, the Livestock farming sector of India grew at a compound annual growth rate of 7.93%, increasing its contribution in agriculture and allied sector from 24.32% in 2014–15 to more than 30% in 2020–21. It also formed nearly 5% of total gross value added in 2020–21. Owing to religious connotations, bovines are a part of Indian culture and India boasts over 300 million cattles, making it the largest bovine population on earth. Consequently, this makes India the number one ranked producer of milk in the world, giving livelihood to 8 crore farmers (PIB, 2023). India now accounts for

over a quarter of global milk production (PIB, 2025a). Likewise, India also ranks 3rd and 8th in global egg and meat industries respectively (PIB, 2023).

Given below is a map that illustrates global livestock population distribution, 2010.



Data: (Gilbert et. al, 2018) Illustration: (Symington, 2023)

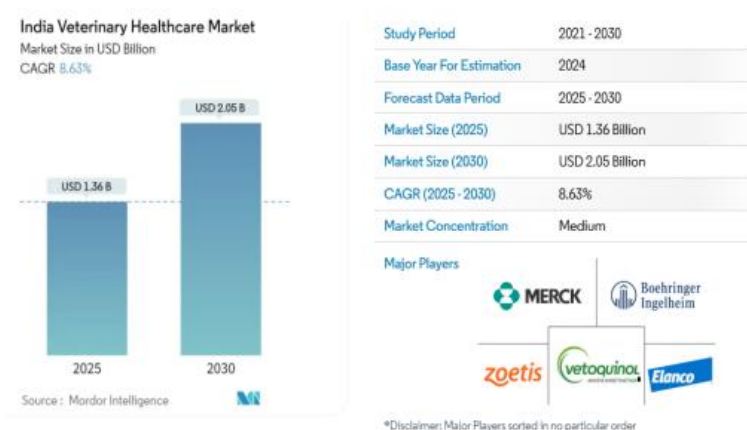
The subject matter of this paper demands a special focus on marginalized farmers and so it becomes important to understand who they are and where they come from. According to the government, a farmer owning less than 1 hectare of land is considered marginalized (PIB, 2019). However, the term „marginalized“ has multiple connotations and for a surface level understanding, it refers to any group or classification which denotes a lack; of resources, of education and awareness, of skill or of access. Marginalization in terms of farmers in India can refer to landless or very less land owning farmers, sharecroppers, tribal farmers, seasonal farm workers or lower caste farmers. Livestock farming can be called the heartbeat of marginalized farmers because for them, the importance of livestock rearing is immense.

Apart from contribution to GDP and a source to livelihood, it provides them with fulfilling access to animal based proteins, calcium and other minerals which help battle

malnutrition and child mortality and would cost much higher if they were to purchase it from markets. Moreover, the surplus produce (meat, milk, eggs etc) can also be used to export, providing an additional source of income. Other than that, livestock rearing also brings along by-products like manure and biogas and acts as a safety cushion against the failure of crops. For landless or livestock-less farmers, the skill to breed and rear animals secures a regular salary from the livestock owners.

A large part of the process of livestock agriculture involves veterinary healthcare. Veterinary healthcare refers to a branch of medicine which deals with prevention, diagnosis and treatment of diseases in animals, both wild and domestic. It is an ever evolving field and is pivotal for animal rearing so much so that it traces its initial practice as far back as the Neolithic age (2400 BC—3000 BC) (Rozzi and Froment, 2018). According to The National Research Council (US) Institute for Laboratory Animal Research, veterinary healthcare also includes transportation, separation, quarantine, surgery and even euthanasia for animals (NRC(US)ILAR, 1996).

The veterinary market in India has lately seen high growth trends. The veterinary healthcare market in India currently stands at \$1.36 billion and is expected to reach \$2.05 billion by 2030 with a CAGR of 8.63% (MordorIntelligence, 2023).



(MordorIntelligence, 2023)

This growth, however, is not uni-dimensional and involves enhanced government accountability, diversified and new research on the subject and upward succession in gaining autonomy of medicine supply indigenously. This is happening due to the collaborative efforts of both, indigenous investments and government. For instance, in October 2022, Indian Immunologicals Limited (IIL) which is an Indian company based in Hyderabad announced that they planned on investing around Rs 700 Crores in setting up an animal vaccine manufacturing facility in the Genome Valley, Hyderabad (also called the ‘Vaccine Hub of the World’ sometimes) to meet the growing necessity of vaccine security in India against economically challenging diseases like Foot and Mouth disease (FMD) and Peste des Petits Ruminants (PPR) and that this might produce nearly 750 new employment opportunities (MordorIntelligence, 2023). Developmental changes like this are important for the entire livestock industry of India and although they originate in cities for business purposes, their impact reaches marginal farmers and rural India too.

Lastly, although the accountability for the welfare of livestock agriculture and marginalized farmers belongs to the government, there are also examples which suggest that wonders can be achieved through public responsibility. These will be discussed in the case studies“ chapter. Veterinary healthcare, without question, is the most significant segment of livestock agriculture. It is responsible for keeping the animals alive and healthy which in-turn assists the farmers in maintaining income stability and producing more for themselves and the nation. The next chapter will talk about the barriers in the growth of livestock agriculture and their dire consequences for farmers and the country, if not checked in time.

Understanding the Barriers in Growth of Marginalized Livestock Farmers

The barriers to livestock farming and veterinary healthcare can be generally characterized under subheadings like economic barriers, infrastructural barriers, institutional

barriers, informational barriers and cultural barriers. These barriers are very significant in understanding the ground reality of marginalized livestock farmers. It has been established thus far that most of these marginalized farmers belong to low income groups and so, are economically weak. The cost associated with veterinary healthcare may take a toll on their pockets. This includes direct cost in form of medicines and veterinarian fees and indirect cost in form of post treatment care, travel cost and opportunity cost of a day's work lost. Many times, poor and marginalized are also unaware of facilities and helpful policies provided by government's banks for getting loans and in desperate times, often borrow money from local moneylenders which is either given at a high interest rate and where inability to pay back might mean mistreatment or even ostracism from the community. In fact, credit is one the primary financial constraints for small, marginalized farmers. The institutional mechanism to protect livestock against credit is weak and despite its importance and demand as an industry, the credit share of livestock farming in total agricultural credit has stayed below 4%. Presently, just 6% of the animal heads (excluding poultry) are covered with insurance (Civildaily, 2017). This keeps farmers at the brink of risk and they do not try to learn or implement new techniques to maintain income stability.

Likewise, infrastructural barriers impose hurdles to marginalized farmers too. This happens due to lack of convenience in travel owing to less public transport options available and terrible roads in many rural set-ups. These logistical barriers do not just increase the travel time and distance to operational clinics but also hamper the timely supply of vaccines and other medicines for animals. Infrastructural barriers also include unavailability or difficult access to cold storage for vaccines in rural areas, unawareness in poor and uneducated farmers about the import- export markets for earning profits and even unregulated slaughter houses. According to a report, almost half of the meat in consumer markets of India comes from unregistered slaughter houses (Pinjari, 2022). Apart from the physical barriers,

there are also mental and societal barriers that come in the way of progress of livestock agriculture. There are many deep seated cultural societal norms that sometimes restrict the access to veterinary healthcare. Some of them include using local superstitious beliefs and rural DIY methods to treat animals instead of seeking professional help, gendered norms that restrict women from going out of their homes, lower castes having to resort to menial resources because they need permission from people belonging to upper caste and so on. Sometimes, many lower caste, tribals and resourceless farmers are unable to voice their concerns due to language barriers between them and professional veterinarians. This specific set of barriers can be classified under both informational and cultural barriers.

It is also concerning that due to rapid urbanization, there is a potential threat about the lack of fodder available to raise animals. This further deepens the challenge in terms of productivity, where India already lacks behind. The average milk production of one Indian cattle unit is roughly 1172 liters per year, about half the global average (Civildaily, 2017). It has already been established that India leads the world in milk production. If productivity can be enhanced, the additional supply can help battle price rise of milk and related products and can even be used for exports. Currently, only 5% or less cropped area in the country is being used for fodder. According to ICAR-Indian Grassland and Fodder Research Institute (IGFRI), Indian deficit in fodder is 11% in dried fodder, 35% in green fodder and 28% in concentrated feed (Pinjari, 2022).

Lastly, the last major barrier that restricts the growth of livestock agriculture for marginalized farmers is the institutional or systemic barrier. This includes issues like lack of professional veterinarian doctors, corruption in the healthcare industry which may include practices like over the counter prescriptions, bias, primary importance to rich farmers etc. It also includes either unavailability or improper access of modern technology to these marginalized farmers that may help them in maximizing their produce and minimizing their

losses. These issues are not important to understand and eradicate only because they hamper economic prosperity of farmers or communities. They are also hazardous to life. In a study, it was found that 60% of animal diseases are zoonotic, which means that it can be spread between animals and humans (Sharma et al., 2022) so in a case of neglect towards livestock, the lives of farmers and even consumers of livestock products are at stake. Moreover, the Veterinary professional to cattle ratio has been recommended by World Organization for Animal Health (OIE) to be at least 1 vet for 500 cattle (Sharma, 2025) but in many regions of India, the situation is terrible, with the state of Jharkhand ranking lowest with 1 veterinarian for around 15704 cattle units (Team Pashudhan Praharee, 2022). Many times, this bad veterinarian professionals to cattle unit ratio is responsible for long waiting queues, less time awarded to individual cases, bureaucratic formalities or mis-timed availability of medicines. Most of these clinics are male dominated spaces, making it difficult for women from marginalized backgrounds to get comfortable and deal with them. This crunch of available vet professionals can lead unaware and poor marginalized farmers to resort to self proclaimed village babas who may fraudulently claim to treat animals and rip these helpless farmers off their pockets.

Lack of veterinary professionals can cause additional problems when geographical hurdles come in way. For example, in mountain states like Jammu and Kashmir where land is mostly unfit for farming, around 80% of households own livestock. Sheep and goats are the most commonly reared animals in the region and it contributes immensely to national wool and mutton supply. That being said, it is also noteworthy that the state only has 1200 veterinarians for nearly 13 million livestock units, an alarming disparity that gives birth to multiple lethal diseases, hampers the growth of families involved and even the awareness of preventive measures (Sharma, 2025). Another aspect of the problem is the fear of lack of growth and incentives among professional veterinary doctors. Many of them prefer cities and

do not like to work in remote and rural areas due to lack of scope, exposure and financial opportunities.

Unfortunately, most of the hindrances in veterinary healthcare are experienced by marginal farmers in rural areas and although there are many efforts undergoing for the betterment of the situation, a lot more needs to be done. Many times, even genuine government and SHGs' efforts are rendered useless due to illiteracy among farmers or poor implementation of schemes by the government which may include extensive paperwork, corruption in bureaucracy, false checks and on-paper approval of situations that might be terrible in real life. Many such schemes require active farmer participation, conveyance access, literacy and on top of all, free time and so, these requirements automatically exclude many marginalized households.

Fortunately, most of these barriers can be overcome through planning and dedication, suggestions relating to which will be provided in the final chapter. The following sub-chapter will elaborate on the petrifying consequences of these barriers, if they persist.

Dire Consequences of these Barriers if Neglected

1. Barriers → Delayed / No Treatment → Disease / Mortality → Income Loss → Lower Resilience.
2. Barriers → Delayed or No Veterinary Care → Higher Disease Burden / Mortality → Loss of Income & Assets → Reduced Household Resilience → Greater Vulnerability to Shocks

The 2 flowcharts illustrated above map out the consequences in case of neglecting or ignoring the barriers. The consequences of ignoring, trying to self-treat or being unable to afford or access healthcare for livestock are multi-faceted and do not just affect individual

farmer families and livestock but also broader livestock resilience, the community and even the country at large. Affecting the herd's health, the negligence and incompetence of the authorities or farmers' neglect towards prevention and early symptoms can cause outbreak of lethal diseases like anthrax, foot-and-mouth disease, brucellosis, rabies, Johne's disease among other animals and even humans. The fear of spread of diseases like the ones mentioned above or any parasitic infection is a constant sword hanging over the head of farmers. They are hazardous to the economy as much as they are to life. For instance, a report by the Indian Council of Agricultural Research (ICAR) suggested that FMD (Foot-and-mouth disease) alone incurs a loss of over Rs 23,000 crore to the country's economy every year (Sharma, 2025). This may cause high mortality rates among the livestock which would in turn affect the quality and quantity of dairy, poultry and meat production. This potentially threatens the family income and would also negatively affect buffer stock and savings, taking a toll on household economies, health and even resilience against unwarranted dangers and emergencies. Many times in situations like this, farmers have to resort to desperate measures like stress selling of animals or reducing the nutritional value in food for livestock. Ultimately, a marginalized farmer going through this can get trapped in a vicious cycle of high debt, low productivity and low income.

Moreover, this situation of dismay might also lead them to giving up which can be manifested in the form of migration from their villages, trying to find alternate sources of livelihood or in some unfortunate cases, even suicides. Farmers are one of the worst suicide affected groups in the country. According to the National Crime Record Bureau, over 1,12,000 farmers took their own lives in India between 2014 and 2024. In fact in 2022 alone, almost 11,500 farmers and farm workers committed suicides. It is important to keep in mind that many such cases are never reported. Also, a report by NSO published in 2018 also suggested that nearly half of the agriculture based families in the country were perpetually in

debt (Fleck, 2024). Marginalized farmers in rural areas of India are also amongst the most vulnerable working population of the country. They are always prone to taking the most damage in case of unforeseen circumstances like drought, market crash, floods or inflation. Moreover, due to bad agricultural produce, many marginal farmer families are forced to take their children out of school to add a helping hand to family income. This deprives many young and deserving minds to gain education and elevate their families from this endless cycle of debt, profit and loss.

The worst hit is taken by marginalized farmers in rural areas because they lack money and support and it is precisely for them that the government has and should dedicate most schemes to. Livestock agriculture in India is a source of livelihood for millions of marginalized farmer families and hence, ensuring its smooth functioning, wellbeing and productivity is pivotal for achieving food security, battling poverty and enabling sustainable development in the villages of the country. These issues require urgent attention and only then India can help its livestock farmers in achieving economic stability while also contributing to the strength and resilience of the rural economy. After becoming aware about the barriers that hamper livestock farming and their possible and dire ill-effects for marginalized farmers, it also becomes important to understand that gradually, the situation is getting better. The next section talks about various policies and schemes that highlight the involvement of the government in the livestock based agriculture industry of India.

The Role of the Government

Goes without saying, the role of government in resolving the barriers that exist in livestock agriculture, elevating the status of marginalized farmers and freeing them from the shackles of continuous misery in the form of livestock mortality rate, vicious cycle of loans and debt and poverty is of paramount significance. It has been the „primary“ agenda of every

government (or has at least been projected) that has been formed in the country to make life better for farmers and so, many schemes and policies have been brought by various governments to improve the livestock agriculture industry of India. This chapter will talk about some of the most important ones, highlighting the role the government has played in trying to overcome the barriers that affect marginalized farmers negatively. These include protection and prevention against lethal livestock diseases, developing livestock infrastructure, building storage for animal produce etc.

First and foremost, the most important and monumental task undertaken by the government in this regard is the Livestock census and integrated animal surveys. A livestock census mirrors the (people) census of India, providing states and district wise gender based census of all the livestock units in the country. The first livestock census was conducted in 1919 and so far, 21 livestock censuses have been conducted (DAHD, 2025a). The primary objective of these censuses is to identify the population and diversity of livestock units, estimate and then provide vaccines, professionals, hospitals, dispensaries, fodder, subsidies and every other element required in the process. A regular livestock census indicates a healthy accountability held by the government, a responsibility for which the marginalized have someone to hold accountable.

One such developmental program for improvements in the livestock industry of India was adopted by the government of India in September 2019 and was a flagship scheme called National Animal Disease Control Programme (NADCP). Under this, the government sought to achieve 100% vaccination in bovines, goats and pigs against FMD (Foot and Mouth Disease) and Brucellosis in five years between 2019–20 to 2023–24 and complete eradication of these by 2030 with a total released amount of rupees 13, 343 crores. This was an all central scheme where 100% needed amount was distributed to all the states and UTs by the central government of India (DAHD, 2025b). The scheme was recently revised by the central cabinet

with an outlay of rupees 3,880 crores for 2024–25 and 2025–26. This revision of the scheme has 3 main elements, National Animal Disease Control Programme (NADCP), LH&DC (Livestock Health and Disease Control) and Pashu Aushadhi. LH&DC further includes Critical Animal Disease Control Programme (CADCP), Establishment and Strengthening of existing Veterinary Hospitals and Dispensaries—Mobile Veterinary Unit (ESVHD-MVU) and Assistance to States for Control of Animal Diseases (ASCAD). An added provision of rupees 75 crore to provide good quality medicines under Pashu Aushadi (animal medicine) component (PIB, 2025b).

Another scheme of similar nature, probably one of the largest and major active scheme run by the government for livestock welfare is the Livestock Health & Disease Control scheme which seeks to improve the quality and quantity of animal produce and lower livestock mortality rates by boosting livestock health through proper medication and vaccination, disease prevention and eradication, opening up of markets and capacity building. Three primary objectives of this scheme were “To implement the Critical Animal Disease Control Programme to eradicate PPR by 2030 by vaccinating all the sheep and goats and to control Classical Swine Fever (CSF) by vaccinating the entire pig population”, to provide doorstep vet facilities to farmers through Mobile Veterinary Units (MVUs) and “To assist States / UTs for Control of Animal Disease (ASCAD) by prevention & control of important livestock and poultry diseases prevalent in different States / UTs as per the States’/ UT’s priorities” (DAHD, 2025c). This would reduce the financial burden on livestock farmers and reduce the risk of humans catching the disease.

A press release by the Ministry of Fisheries, Animal Husbandry & Dairying in 2023 recalled various schemes undertaken for animal welfare between 2014–15 to 2022–23. Some of these include Rashtriya Gokul Mission launched in December 2014 to conserve indigenous species of Bovines through steps like Artificial Insemination Programme, Sex sorted semen

production, Breed Multiplication Farms, Breed Improvement programme using IVF, establishment of MAITRIs (Multi-Purpose AI technicians in Rural India), e-GOPALA app and countless such measures; The National Programme for Dairy Development (NPDD) in February 2014 to improve the state of Milk farming in India. This includes “strengthening of infrastructure for production of quality milk, Procurement, Processing and Marketing of Milk & Milk Products by the State Implementing Agency (SIA)”; Dairy Processing and Infrastructure Development Fund (DIDF) in December 2017 to help modernize the “milk processing, chilling and value addition infrastructure” for the Dairy Co-operative, Multi State Dairy Cooperative, Milk Producer Companies (MPC), NDDBs subsidiaries. It helped create Milk Processing Capacity of 63.70 Lakh Liter per Day (LLPD), Chilling capacity (Bulk Milk Cooler (BMC)) of 3.4 LLPD, Milk Drying capacity of 265 Metric Tonne per Day (MTPD) and Value Added Products (VAP) manufacturing capacity of 10.46 LLPD (PIB, 2023). These are only a select few schemes and policies out of the sea of actions taken by the government, all of which can not find a mention here. In addition to these schemes dedicated to particular causes, the government also provides farmers with information through Focused Publicity Campaigns, Kisan Call Centres (KCCs), Agri—Clinics and Agri-Business Centres (ACABC) of entrepreneurs, Agri Fairs and exhibitions, Kisan SMS Portal etc.

The developments have been done keeping in mind the topography and diversity of the country. For instance, In Jammu and Kashmir, initiatives like subsidized fodder distribution and mobile veterinary camps have been implemented. Although still struggling to perform as expected, this has shown promise because of its resilient thought and strong willpower of all the workers toiling day in and out to bring the country’s remotest and most conflict-affected areas to the mainstream (Sharma, 2025). Moreover, the government has also been very generous in allocating funds for tasks which hints towards a genuine effort to revolutionize the livestock industry of India. For example, Infrastructural development funds

like Animal Husbandry Infrastructure Development Fund was launched with a budget of rupees 15000 crore for investment to companies, farmers, etc. to set up meat, dairy processing and Animal Feed Plants. Likewise, to modernize the best in the world dairy industry, rupees 8004 crores were invested in Dairy Processing and infrastructure Development Fund (Pinjari, 2022).

Both the states and central governments also stay alarmed during the times of crisis, working together to achieve sustainable standards in livestock farming and putting the poor and marginalized farmers out of their misery in the present and for the future. For example, during the initial outbreak of the Covid-19 in March of 2020, the Government of India firmly stated that all the states of the union must ensure that medical help for animals should be treated as an essential service meaning it should not get suspended during COVID-19 lockdowns across the country (MordorIntelligence, 2023). Likewise, in July 2022, the Union Minister for Fisheries, Animal Husbandry, and Dairy, Shri Parshottam Rupala, strongly recommended that the cooperative movement in the livestock industry should be analyzed and strengthened (MordorIntelligence, 2023). Therefore it is safe to say that the situation throughout the country is getting better, with governments showing more interest in veterinary healthcare and raising awareness for the same. A good growth in the sector can be expected in years to come. However, even these well thought and extravagant efforts in forms of policies and schemes do not come without their own issues and challenges. There is a visible gap that persists between framing and implementation of beneficial schemes. To give an example, despite efforts to modernize the animal healthcare in rural India, many veterinary centres in the country are still understaffed, located at very remote places from many villages without having convenient means of public transport, have old technology and are riddled with corruption (Sharma, 2025). The efforts of the government and all the technology in the world would not be able to help marginalized farmers in rural India unless changes occur at

grassroot level and rules are implemented seriously. Some of these suggestions to improve the current state of affairs are provided in the last chapter.

The next chapter will discuss a few case studies to bring the research in the purview of practical life.

Case studies

After gaining a comprehensive familiarity with the basic functioning, issues and their possible consequences and other important components of livestock agriculture in India, it becomes important for the reader to also know how these elements play out in practical situations and for this purpose, the paper would follow the discussion thus far with two real life examples and stories that might help in achieving a better understanding of how certain factors influence marginalized farmers involved in livestock agriculture. The two case studies are very different, depicting two sides of the same story. First case study is situated in Maharashtra and talks to a few small farmers who lost their livestock to LSD (Lumpy Skin Disease). It also brings in the perspective of a veterinarian and then situates these particular cases in a larger Indian context. The other case study talks about how a collaboration between professional veterinarians and dedicated community animal health workers (CAHW) support groups can help the terrible status of animal welfare in the country, depicting a perfect example of how public will and professional effort can help the country deal with many issues at hand. The two case studies, although not directly linked, can be read as connected through the fabric of betterment of livestock health.

In 2022, India experienced what can be called an LSD (Lumpy Skin Disease) epidemic. LSD is a viral infection found specifically among cattle and is spread through blood transmitting insects like ticks and mosquitoes. It can cause nodules and swelling of organs, fever, loss of appetite, eye and nasal discharge, weight loss and can trigger reduced

milk production in cows and or can even cause painful death. Due to its widespread infection in 2022, it claimed the lives of nearly 3 million cattle units in India across 23 states, with estimates suggesting almost 97,000 bovines died between just July and September. Although the exact extent of loss can not be portrayed, it was estimated by United Nations“ Food and Agriculture Organization“s 2020 risk assessment report that Lumpy caused an economic loss of \$1.45 billion across south east Asia (Jain, 2023). This unfortunate outbreak also caused many farmers to evacuate and search for other sources of income.

This particular group of case studies are situated in Maharashtra. A middle aged couple, Pandurang Khondre and his wife Malan became one of the many victims of this epidemic when they observed a lack of appetite and few red colored nodes on the right leg of their strongest Ox, Khandya. They had to spend over 16 hours everyday to take care of him and even consulted a private veterinarian because “The Government doctors wouldn’t show up. There were times we waited for an entire day”, they said. Khandya could not be saved even after the farmer couple spent more than rupees 60,000 for over 3 months for his treatment. Ultimately, Khondre had to get another Ox for rupees 15,500 or more. Loss of a working unit nearly halved Khondre“s income and made him worried about a stable source of income (Jain, 2023). To keep in mind, this was an additional problem alongside the already existing issue like climate change, floods, heatwaves and other factors that impact such small scale farmers.

Similarly, a 70-year-old Maharashtrian farmer, Vitthal Kumbhar recalled when one of his cows and a bull calf got struck by Lumpy. It took over two months for both the animals to recover and during this time, Kumbhar“s son transported each of the four public veterinarians they consulted back and forth on his bike every day for the 90 injections they provided in a month. The milk production from the lumpy affected cattle decreased post treatment and the family had to purchase milk from the market to feed the calves, eating up a sizable portion of

their income. In October same year, Kumbhar and his family also lost the majority of their sugarcane crop to hailstorm. They could not even recover the costs and had to get into the vicious cycle of getting loans to repay loans (Jain, 2023). Earning a profit seems like a far and unreachable prospect to such families when they cannot even mitigate a loss without getting a loan, repaid using another loan.

The governments and authorities acted swiftly and as of February 2023, 84.19 million cattle had been vaccinated against LSD. Jitendra Kurundwade, assistant commissioner of Kolhapur's animal husbandry department, said that sometimes there were cases where they were treating the same cattle for a month and so, acting in time, the authorities de-centralized the vaccination process and vaccinated all the bovines in the region. Through this, the vaccination process that could have taken over 6 months to finish, culminated in weeks. Likewise, Raosaheb Salunkhe, a public veterinarian in Kolhapur district said that during the peak of the outbreak, they were sometimes treating 80 cases in a single day because farmers usually came to them after consulting private vets and the situation becoming severe (Jain, 2023).

The next case study can be read as a solution to the above mentioned incidents. Pashu Sakhis, literally translating into "friends (female) of animals", is a non-governmental and voluntary group of women working in the fields and farms of rural India, doing their part in spreading awareness and mitigating the lack of professional vets available. These women are trained animal health service providers who are not just nurses that nurture poultry and cattle but also micro-entrepreneurs and trailblazers that have initiated grassroots changes. Part of their purpose is also to learn and teach how livestock can be managed sustainably to generate maximum output at minimum cost. Many of these sakhis are trained under multiple programs organized by CSRs (Corporate Social Responsibility), NRLMs (National Rural Livelihoods Mission) and other developmental organizations to mitigate the lack of professional

veterinary access and availability in rural areas (Global Services TGT, 2025). Ms Mira Devi, a community animal health worker in Bihar said that her work has helped a lot of farmers in the village community who now trust her. She even works with the village administration or „Gram Panchayat“ to eradicate many local issues (WOAH, 2025). This case study is a prime example of how many problems faced by rural India can be helped by the support of educated and resourceful citizens. One can only wonder how good things can be if the actual authorities in power dedicated themselves completely to the upliftment of marginalized livestock farmers. The next and final chapter will conclude the gist of this paper and provide some well thought suggestions that may help battle the problems that persist for marginalized farmers in the livestock agriculture industry of India.

Suggestions: The Way Forward

It would not be correct to say that there have been no improvements in livestock agriculture for the marginalized farmers of India. In fact, there has been a complete turn - around in perhaps the last fifty years about the way policy decisions are formed, technology is used and market works. But the questions since then have also risen; how many of these changes have actually been implemented? Is the current rate of development in livestock agriculture enough to keep up with the growing needs of farmers and consumers? Is the system machinery capable enough to deal with or provide a safety cushion against calamities and unforeseen circumstances? These questions are situated in practical situations and can not be answered in yes or no. However, having read and analyzed the subject matter at length, it is safe to say that although a lot has improved, much still faces neglect and this section will try to offer some resolutions and point out the research gap to work on for future researchers to make the situation better.

One of the most significant areas for improvement is working on ease of access to livestock healthcare for small farmers. Lack of veterinary professionals in rural areas leads to either unavailability of quality healthcare or increased expenditure to reach nearest government hospitals. Many healthcare professionals avoid going to rural areas because they feel city life can provide them with more opportunities. For such healthcare workers who lose out on opportunities of city life, governments should provide added incentives. The institutional framework of veterinary healthcare should also be strengthened by opening up more colleges. Currently, the country only has 56 recognized veterinary colleges (MFAHD, 2023). More colleges in smaller and less developed regions would also lead to more opportunities for students belonging to marginalized communities to become vets themselves. This would elevate the community because these vets would overcome cultural and linguistic barriers which a vet from outside the region might probably encounter. The goal of providing affordable and quality healthcare becomes more difficult in locations with difficult accessibility. India is a country with diverse landforms and in multiple areas, harsh weather conditions account for high mortality rates among livestock units. It is challenging for accessible healthcare to reach timely in regions with mountainous or desert terrains and hence, it should be the primary agenda of the authorities to make conveyance easy in these regions. Other than that, attendance of vet professionals should be regularly monitored and regular defaulters should be penalized.

Another area where a lot of improvement and development needs to be initiated is technological advancements. Setting up and expanding the usage of technology to help marginalized farmers is a two step work. It does not stop with the government introducing new policies or advanced methods of farming. The benefits of technology can only be reaped when it reaches common masses like marginalized rural farmers. Authorities should make sure that the awareness programmes like helpline numbers, mobile veterinarian chatbots and

on-site help providers are functional and implemented. Many times, issues of farmers become grave because either the farmers are not aware about prevention or first aid to save the moment. Some other times, many unaware marginalized farmers can fall prey to blood sucking village loan-givers or local „quacks“ who may knowingly or unknowingly provide them with advice that can negatively affect their livelihood. To explain the gravity of the situation better, reports suggest that only about 5% of the farm households in India access information on livestock technology (civildaily, 2017). This is a very small percentage and goes without saying, needs urgent intervention. Awareness can also be raised through farmer training collectives and organizing village fairs. Efforts should also be put in the research on technology that deals with genetic surveillance, especially the disease and death causing virus so that the pattern can be analyzed and better preventive care can be taken. Genetic research can also be done in collaboration with research institutes so as to locate and preserve high yielding and more resilient to climate and disease genes in an indigenous gene bank. Thus far, only limited success has been achieved in this area of artificial insemination and cross breeding for improving productivity. For instance, even after over three decades of crossbreeding, the crossbred population in India by 2017 was only 16.6% in cattle, 21.5% in pigs and 5.2% in sheep (civildaily, 2017).

Higher focus should be given to a better network of ambulances because for rural marginalized farmers, the majority of the issues arise because they do not have viable means to reach the nearest government veterinary hospital. There should also be compulsory vaccination for all livestock units because spreading of fatal disease can be lethal to the entire ecosystem of agriculture. A prime example of this, as discussed in the previous chapter, is the lumpy disease break out of 2022. Since the subject matter deals with rural Indian farmers, it goes without saying that apart from providing technical assistance and raising awareness, the focus should majorly be on the village communities which would include making healthcare

inclusive and affordable. For this, special initiatives should be taken to strengthen women self-help groups and tribal communities rearing animals. According to a study conducted in Bangladesh, women-led livestock farming is linked to rural community development, women's rights upliftment, better child nourishment, increased family income and hence, strong nation building too (Nath et al., 2024). Lastly, development of livestock agriculture should be initiated that would include mixing up animal husbandries and poultry farming with employment schemes like MNREGA to help landless and small farmers.

The government should focus more on commercialization of markets, opening them up for the farmers to help them find the highest suitable prices. Most of the animal based products like milk and meat have a short shelf life and so many times, the farmers are forced to sell their produce at lower prices to big corporate houses. Moreover, just like efforts are put in by the government to achieve food security, it is also very important to achieve fodder security. Fodder is the fuel of the livestock industry. According to a report by the Ministry of New and Renewable Energy (MNRE), India produces over 500 million tonnes of crop remains every year out of which almost 100 million tonnes are burnt to ground (drishtias, 2022). This can potentially be used as fodder for cattle and goats. There have been scientific developments in this area that have generated high yielding varieties of fodder which can be used for better results. The government should dedicate some effort into more R&D in this sector and implement its usage.

Last but not least, it is also a civic duty of citizens to get inspired by CAWG (Community Animal Welfare Groups) initiatives like Pashu sakhis¹, come together and contribute or collaborate with governments if they can to help eradicate, prevent and raise awareness about the issues of livestock farmers belonging to marginalized groups.

Case Study

- Village: Tikri—middle sized village in sector 48, Gurgaon
- Sizeable population with a large range of economic stability amongst different households.
- Many household own livestock, including Cows, Buffalo, Goat herds. Less keep chickens
- Of the 27 people interviewed, many rely on milk sales as their primary source of income. Few run a supply business to the main city, and others also own farmland and sell crops.

People in household	Main income source	Number of Cows kept	Number of Buffalo's kept	Private or Government Care	Time taken for vet to arrive at house/ min	Vet for regular checkups?	No. of times vet called per year	Difficulties faced in accessing vet care	
								Financial Burden	Unavailable vets
11	Milk Sales	1	2	Private	15	No			
9	Milk Sales	5	20	Government	15	No	4-5 per month		
10	Farm and Milk Sales	11	1		15	No	every 20 days doctor is needed. Doctors only called when sick.		
4	Milk Sales	1	0	Private	15		2-3 months ago		
4	Milk Sales	2	0		15		2/month in monsoon.		
2	Milk sales driver employment	1	5	Private	by appointment time	No	only when sick	Government and Private	Government
10	Business, milk sales, Farm	2	1	Private	15	No	called only when animals sick	Government and Private	Government
4	Milk sales, Farm			Private	15	No	Only called when animal sick	Government and Private	Government
1 (temple residence)		3	0	Private	30		only called when animal sick	Government and Private	Government
9	Milk sale corporate job			Private	60+	No	1	Government and Private	Government
5	Business	4	2	Private	30	No	last week. called only when ill	Government and Private	Government
12	Milk Supply to gurgaon	4	4	Private	Appointment basis	No	at least 1/ month the vet is called due to illness. otherwise doctor is called if illness occurs		Government
5	Milk sale		1	Private	Appointment basis	No	only called when major illness occurs	Private	Government
7	Goat meat Milk supply	1	2	Government	Appointment basis	No	1-2 times in past year; otherwise only when sick	Private	Private and Government
7	Milk Supply		3	Private	1 hour	No	called 4-5 times per year, only when sick		Private and Government
11			large herd	Private	1 hour	No	doctor called only when sick. most recently called 1 month ago		Government
7	Milk Supply	2	4	Private	Appointment basis	No	2-3 times in past year. doctor called only when sick	Private and Government	Private and Government
17	Milk supply	5	large herd	Both	private - 30min or on appointment basis. Gov- 1 hrs but unreliable	Yes	Doctor called many times per year - large herd falls sick often + upkeep requires constant vet support	Private and Government	Government

11	Milk sales - supply business	4-5	10-12	Private	30	No	Only call for major illnesses	Government and Private	Government
11	Milk Sales Business	1	2	Private	Appointment basis	Yes	called 3-4 times per year minimum. otherwise called when animals ill	Government and Private	Government
8	Employment	0	1	Private	60	No	only called for major illness	Government and Private	Government
	Milk Sales	1	22	Both	Appointment basis	No	Only called for major illnesses	Government and Private	
17	Milk Sales Driving business	2-3	8-10	Private	Appointment basis	No	2-3 times called in past year. Otherwise only when major illness	Private	Government
6	Milk supply	2	0	Private	30	No	doctor called 2 times per year and when other major illness occurs	Private	Government
5	Milk supply	0	1			No	Have never called doctor before - no major illness occurred	Government and Private	
10	Small business	2	5	Private	1-2hrs	No	called 1-2 times in past year. called only when sick		Government
Nuclear family	Dental hygienist - employed	1	2	Both	1-2hrs	No	8-10 times in the year. otherwise only when sick		Private

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